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1  /*****
2  * DELTA ADJUSTED MI + LMM
3  *****/
4  /*****
5  * Set library
6  *****/
7  libname eddy "/home/u64020665";
8
9  /*-----
10 0) Housekeeping (optional)
11 -----*/
12 ods graphics off; /* keep simple; you can turn on if desired */
13
14 /*-----
15 1) Prepare data: missingness indicator for BPRS
16 -----*/
17 data bprs_base;
18     set eddy.alz_long2;
19     missBPRS = (BPRS = .);          /* 1 if missing, else 0 */
20 run;
21
22 /*-----
23 2) Multiple Imputation under MAR for continuous BPRS
24     - FCS regression for BPRS
25     - Include time, time2, baseline covariates
26 -----*/
27 proc mi data=bprs_base out=mi_bprs seed=20251230 nimpute=10;
28     class patid sex;
29     fcs reg(BPRS = time time2 age sex bmi adl abpet0 tauet0);
30     var patid time time2 BPRS missBPRS age sex bmi adl abpet0 tauet0;
31 run;
32
33 /*-----
34 3) Delta-adjustment scenarios (MNAR departures)
35     Interpretation:
36         BPRS_adj = BPRS + delta for rows originally missing (missBPRS=1)
37         BPRS_adj = BPRS          for rows originally observed
38         Choose a brief but informative grid of deltas
39 -----*/
40 data mi_bprs_delta;
41     set mi_bprs;
42
43     /* MAR baseline */
44     delta = 0;
45     BPRS_adj = BPRS + delta*missBPRS;
46     output;
47
48     /* MNAR: missing values higher than MAR predicts */
49     delta = 5;
50     BPRS_adj = BPRS + delta*missBPRS;
51     output;
52
53     delta = 10;
54     BPRS_adj = BPRS + delta*missBPRS;
55     output;

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55
56 /* MNAR: missing values lower than MAR predicts */
57 delta = -5;
58 BPRS_adj = BPRS + delta*missBPRS;
59 output;
60
61 delta = -10;
62 BPRS_adj = BPRS + delta*missBPRS;
63 output;
64 run;
65
66 proc sort data=mi_bprs_delta;
67 by delta _Imputation_ patid time;
68 run;
69
70 /*-----
71 4) Refit the FINAL LMM (Model 1) for each (delta, imputation)
72 - Random intercept + random slope for time
73 - Fixed effects same as your selected model
74 - Save fixed effects (SolutionF) for pooling
75 -----*/
76 ods output SolutionF=fixef_delta;
77
78 proc mixed data=mi_bprs_delta method=reml;
79 by delta _Imputation_;
80 class patid sex;
81
82 model BPRS_adj =
83     time time2
84     age sex bmi adl abpet0 taupet0
85     time*age time*sex time*adl
86     / solution ddfm=kr;
87
88 random intercept time / subject=patid type=un;
89 run;
90 ods output close;
91
92 /*-----
93 5) Pool fixed effects across imputations within each delta
94 - This produces pooled estimates and SEs per Effect per delta
95 -----*/
96 /* Keep only effects we want to track across delta scenarios */
97 data fixef_delta_keep;
98 set fixef_delta;
99 where Effect in ("time", "time2", "age", "bmi", "adl", "abpet0", "taupet0");
100 run;
101
102 proc sort data=fixef_delta_keep;
103 by delta Effect;
104 run;
105
106 proc mianalyze data=fixef_delta_keep;
107 by delta Effect;
108 modeleffects Estimate;
109 stderr StdErr;

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110 run;
111
112 ods output ParameterEstimates=pooled_pe;
113
114 proc mianalyze data=fixef_delta_keep;
115     by delta Effect;
116     modeleffects Estimate;
117     stderr StdErr;
118 run;
119
120 ods output close;
121
122 data pooled_long;
123     set pooled_pe;
124
125     /* Keep only what we need */
126     keep delta Effect Estimate StdErr LowerCL UpperCL Probt;
127 run;
128
129 proc sort data=pooled_long;
130     by Effect delta;
131 run;
132
133 proc print data=pooled_long noobs;
134 run;
135
136 proc transpose data=pooled_long out=wide_est(drop=_name_) prefix=Est_;
137     by Effect;
138     id delta;
139     var Estimate;
140 run;
141
142 proc transpose data=pooled_long out=wide_p(drop=_name_) prefix=P_;
143     by Effect;
144     id delta;
145     var Probt;
146 run;
147
148 data wide_q6;
149     merge wide_est wide_p;
150     by Effect;
151 run;
152
153 proc print data=wide_q6 noobs;
154 run;
155
156 data q6_long;
157     set wide_q6;
158     length delta 8 estimate 8;
159
160     array ests[5]
161         'Est_-10'n
162         'Est_-5'n
163         Est_0
164         Est_5

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165     Est_10;
166
167     array deltas[5] _temporary_ (-10 -5 0 5 10);
168
169     do i = 1 to 5;
170         delta = deltas[i];
171         estimate = ests[i];
172         output;
173     end;
174
175     keep Effect delta estimate;
176 run;
177
178 /* Keep only time and time2 */
179 data q6_long_tt;
180     set q6_long;
181     where Effect in ("time", "time2");
182 run;
183
184 proc sgplot data=q6_long_tt;
185     series x=delta y=estimate / group=Effect markers;
186     refline 0 / axis=y;
187     xaxis label="Delta (MNAR shift applied to missing BPRS)" integer;
188     yaxis label="Pooled estimate";
189     keylegend / location=inside position=topright;
190 run;
191
192 proc sgpanel data=q6_long_tt;
193     panelby Effect / columns=2;
194     series x=delta y=estimate / markers;
195     refline 0 / axis=y;
196     colaxis label="Delta" integer;
197     rowaxis label="Pooled estimate";
198 run;
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